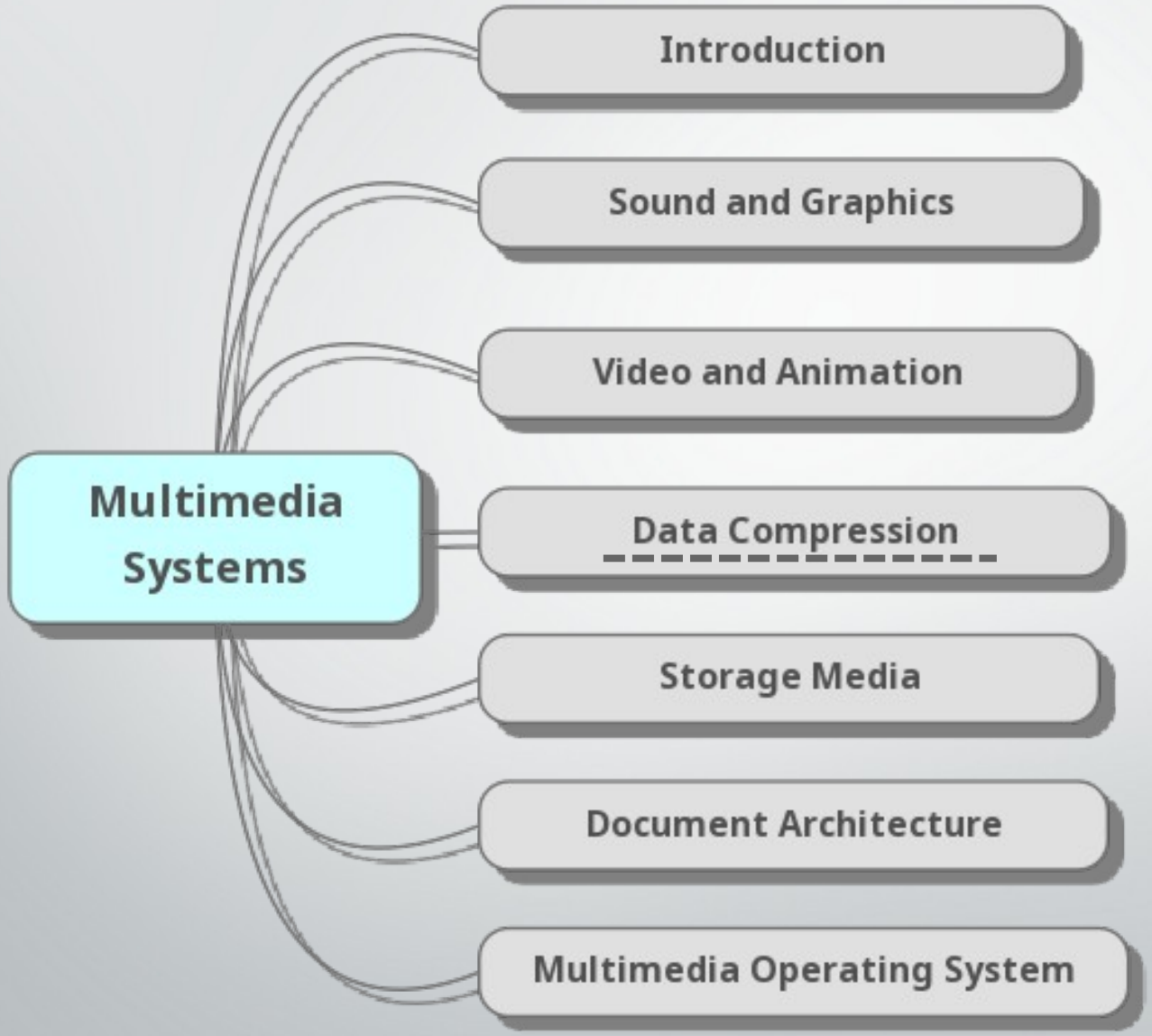
# Multimedia System

[BE IT 5th Semester]

Rishi K. Marseni
{rkmarseni@ncit.edu.np}


Multimedia Systems
Introduction
Sound and Graphics
Video and Animation
Data Compression
Storage Media
Document Architecture
Multimedia Operating System

# 4. Data Compression Methods

**4.1. Coding Requirement**

**4.2. Types of Compression:**
Lossy and LosslessCompression

4.2.1. Run length Coding, Entropy Coding

4.2.2. Huffman Coding

4.2.3. Arithmetic Coding

**4.3. Coding Standards**

4.3.1. JPEG: JPEG compression steps

4.3.2. MPEG: Evolution of MPEG, Data Stream of MPEG,MPEG Audio

**4.4. Digital Video Interactive (DVT), h.261(p✖64)**

## Data Compression and Coding Fundamentals

### Why we need image compression?

- One 90 minutes color movie, each second plays 24 frames. When we digitize it, each frame has 512X512 pixels, each pixel has three components RGB each one occupies 8 bits respectively, the total byte number is:

  90X60X24X 3X512X512 = 97,200MB

- A CD may save 600 megabytes data, the movie needs 160 CDs to save.

# Data Compression and Coding Fundamentals

Objective: To reduce the amount of data required to represent an image.

Important in data storage and transmission

- Progressive transmission of images (internet, www)
- Video coding (HDTV, Teleconferencing)
- Digital libraries and image databases
  - Medical imaging
  - Satellite images

# Data Compression and Coding Fundamentals

## Storage Space and Coding Requirements *Storage Requirement*

Uncompressed graphics, audio and video data require considerable storage capacity.

For e.g. an image represented in 640 by 480 resolutions with 8 bits per pixel require 300 kilobytes of storage space.

So, in order to reduce the storage requirement of the multimedia data it should be compressed.

The compression should be such that there should not be a loss of quality of the data and hence compression should not reduce the information content of the data. For e.g. 90% of the raw audio can be deleted without affecting the quality of the audio.

# Data Compression and Coding Fundamentals

## *Bandwidth Requirement*

Uncompressed data transfer requires greater bandwidth or data rate for its communication.

If the data is compressed there can be a considerable reduction in the bandwidth requirement for the transmission of the data.

# Data Compression and Coding Fundamentals

## *Bandwidth Requirement*

The following examples specify continuous media and derive the amount of storage required for one second of playback:

- An uncompressed audio signal of telephone quality sampled at 8 kHz and quantized with 8 bits per sample requires 64 kbits to store one second of playback and a bandwidth of 64 kbits/sec.
- An uncompressed stereo audio signal of CD quality is sampled at a rate of 44.1 kHz quantized at 3 16 bits per sample require 705.6 * 10 3 bits to store one second of playback and the bandwidth requirement is 705.6 * 10 bits/second.

# Data Compression and Coding Fundamentals

***Bandwidth Requirement***

As mentioned above compression in multimedia systems is subject to certain constraints.

- The **quality** of the compressed data should be as good as possible.
- To make **cost-effective** implementation possible, the complexity of the technique used should be minimal.
- The processing of the **algorithm** must not exceed certain **time span**.

# Data Compression and Coding Fundamentals

***Bandwidth Requirement***

In retrieval mode application, the following demands arise:

- ***Fast forward and backward data retrieval*** should be possible with simultaneous display.
- ***Random access*** to single images and audio frames of a data stream should be possible without extending the access time more than 0.5 second.
- **Decompression** of data should be independent of other data units.

# Data Compression and Coding Fundamentals

***Bandwidth Requirement***

For both dialogue and retrieval mode, the following requirements apply

- The format should be **independent** of frame size and video frame rate.
- The format should support **various** data rate.
- There should be **synchronization** between the ***audio*** and ***video.***
- Compression and decompression should **not require additional hardware.**
- The compression of data in **one system of multimedia** should ensure the **decompression** in the **other system**.

# Data Compression and Coding Fundamentals

## Source, Entropy and Hybrid Coding

### Entropy encoding

- Data stream is considered to be a simple digital sequence without semantics
- Lossless coding, decompression process regenerates the data completely
- Used regardless of the media s specific characteristics
- Examples: **Run-length encoding, Huffman encoding, Arithmetic encoding**

### Source encoding

- Semantics of the data are taken into account
- Lossy coding (encoded data are not identical with original data)
- Degree of compression depends on the data contents
- Examples: Content prediction techniques; e g. use of spatial redundancies between still images for data compression.

  Discrete Cosine Transformation (DCT) as transformation technique of the spatial domain into the two-dimensional frequency domain

### Hybrid encoding

- Used by most multimedia systems
- Combination of entropy and source encoding
- Examples: JPEG, MPEG, H.261, DVI

# Lossy Vs. Lossless Compression

**Compression techniques**

Information preserving

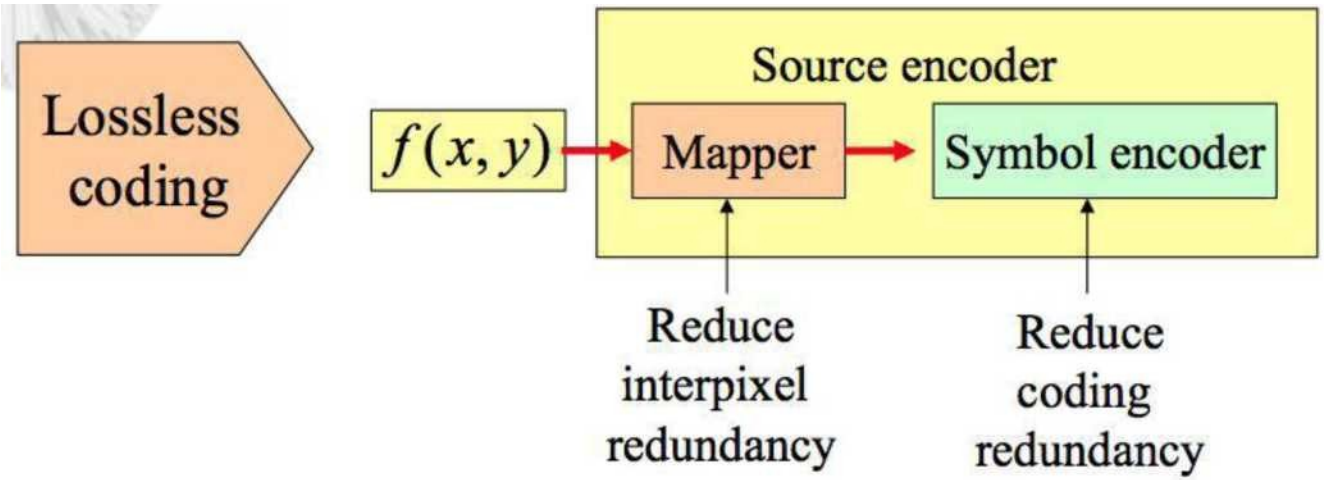
**(loss-less)**

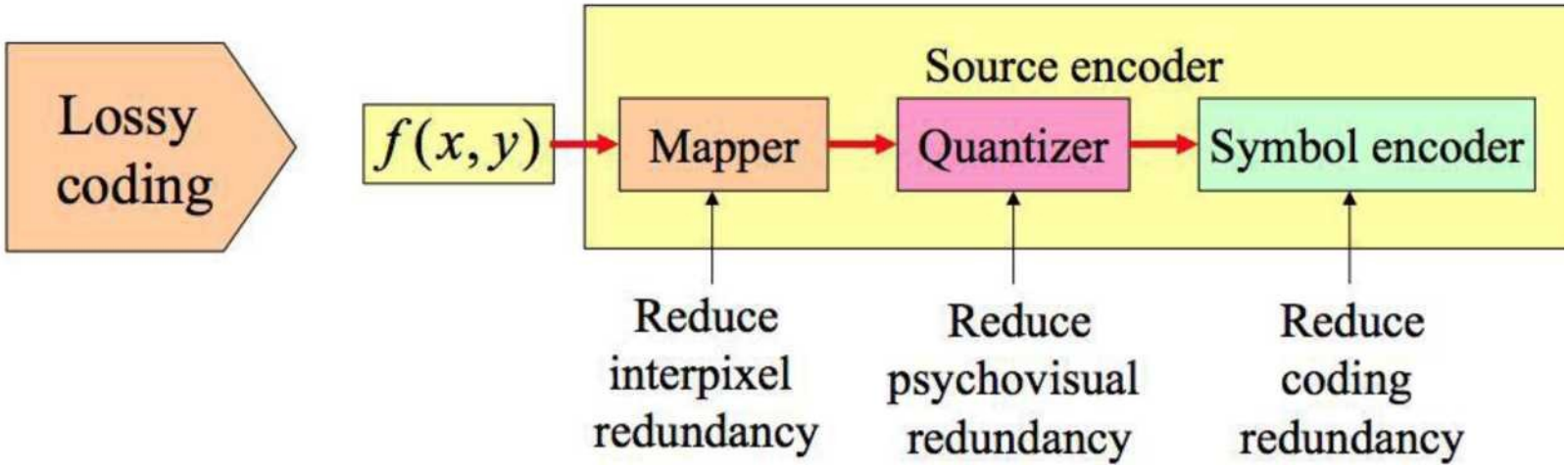
Images can be compressed and restored without any loss of information.
Application: Medical images, GIS

**Lossy**

Perfect recovery is not possible but provides a large data compression
Example : TV signals, teleconferencing

# Lossless Vs. Lossy Coding



# Data Compression and Coding Fundamentals

## Source, Entropy and Hybrid Coding

### Source Coding

Source coding takes into account the semantics and the characteristics of the data. Thus the degree of compression that can be achieved depends on the data contents.

Source coding is a lossy coding process in which there is some loss of information content. For e.g. in case of speech the speech is transformed from the time domain to frequency domain.

# Data Compression and Coding Fundamentals

## Source, Entropy and Hybrid Coding

## Source Coding

In the psychoacoustic the encoder analyzes the incoming audio signals to identify perceptually important information by incorporating several psychoacoustic principles of the human ear.

One is the critical-band spectral analysis, which accounts for the ear"s poorer discrimination in higher frequency regions than in lower-frequency regions. The encoder performs the psychoacoustic analysis based on either a side-chain FFT analysis or the output of the filter bank.

E.g. Differential Pulse Code Modulation, Delta Modulation, Fast Fourier Transform, Discrete Fourier Transform, Sub-band coding etc

# Data Compression and Coding Fundamentals

## Source, Entropy and Hybrid Coding

### Entropy Coding

Entropy coding is used regardless of the media"s specific characteristics.

The data stream to be compressed is considered to be a simple digital sequence and the semantics of the data is ignored. It is concerned solely with how the information is represented.

# Data Compression and Coding Fundamentals

**Source, Entropy and Hybrid Coding**

**Entropy Coding**

***Run Length Coding***

Typical applications of this type of encoding are when the source information comprises long substrings of the same character or binary digit. Instead of transmitting the source string in the form of independent code-words or bits, it is transmitted in the form of a different set of codewords which indicate not only the particular character or bit being transmitted but also indication of the number of characters/bits in the substring.

# Data Compression and Coding Fundamentals

## Source, Entropy and Hybrid Coding

## Entropy Coding

### *Run Length Coding*

For e.g. if the string is AAAAABBBTTTTTTMMMMMMMM, it is encoded as A!5BBBT!6M!8 (In this case there is no point in encoding characters that repeats itself less than 4 times).

*Diatomic encoding* is a variation of run-length encoding based on a combination of two data bytes. This technique determines the most frequently occurring pairs of bytes. For e.g. in English language “E”,”T”,”TH”,”A”,”S”,”RE”,”IN” and “HE” occurs most frequently.

# Data Compression and Coding Fundamentals

## Source, Entropy and Hybrid Coding

## Entropy Coding

***Huffman Coding***

Huffman coding is an example of variable length coding. It is based in the concept that the probability of occurrence of the characters is not same so different number of bits is assigned for different character.

Basically in variable length coding the characters that occur most frequently are assigned fewer numbers of bits. However in order to used variable length coding the destination must know the set of code-words being used by the source. In Huffman Coding the probability of occurrence of the characters are estimated and based on this estimation code-words are assigned to the characters.

## Q.N.1 :-Design Huffman code for probabilities:

## 0.1, 0.4, 0.06, 0.1, 0.04, and 0.3.

- Huffman coding: give the smallest possible number of code symbols per source symbols.

Step 1: Source reduction

**Solved in Class !**

## Step 2: Code assignment procedure

**Solved in Class !**

The code is instantaneous uniquely decodable without referencing succeeding symbols.

**Average length:**

(0.4) (1) + 0.3 (2) + 0.1 X3 + 0.1 X4 + (0.06 + 0.04) 5

= 2.2 bits/symbol

# Q.N.2:-Design Huffman code for probabilities: 0.1, 0.4, 0.15, 0.05, 0.1, and 0.2.

# Q.N.3: let the word to be encoded be AABABBEEEEFEFC

Ans:- Length of string=14

No. of A =3 Probability of occurrence of A=3/14=**0.214**;

No. of B=3 Probability of occurrence of B=3/14=**0.214**

No. of E=5 Probability of occurrence of E=5/14=**0.357**

No. of F=2 Probability of occurrence of F=2/14=**0.142**

No. of C=1 Probability of occurrence of C=1/14=**0.01**

Here , Huffman code for probabilities are :

0.214, 0.214, 0.357, 0.142 and 0.01.

And solve.....

# Q.N.4:- Construct the Huffman code for

| Gray Level | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| No.of Pixel | 5000 | 1000 | 500 | 530 | 1250 | 950 | 860 | 130 |

**Solved in Class !**

# Q.N.5:- Construct the Huffman code for

| Gray Level | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|---|
| No. of Pixel | 5320 | 1000 | 500 | 525 | 1236 | 956 | 856 | 128 |

**Solved in Class !**

# Data Compression and Coding Fundamentals

**Source, Entropy and Hybrid Coding (cont..)**

**Entropy Coding (cont..)**

***Arithmetic Coding***

Unlike Huffman coding which used a separate codeword for each character, arithmetic coding yields a single codeword for each encoded string of characters.

The first step is to divide the numeric range from 0 to 1 into a number of different characters present in the message to be sent - including the termination character - and the size of each segment by the probability of the related character.

# Data Compression and Coding Fundamentals

## Source, Entropy and Hybrid Coding (cont..)

### Hybrid Coding

This type of coding mechanism involve the combine use of both the source coding and the entropy coding for enhancing the compression ratio still preserving the quality of information content. The example of Hybrid Coding includes MPEG, JPEG, H.261, DVI techniques.

# Basic Data Compression Techniques

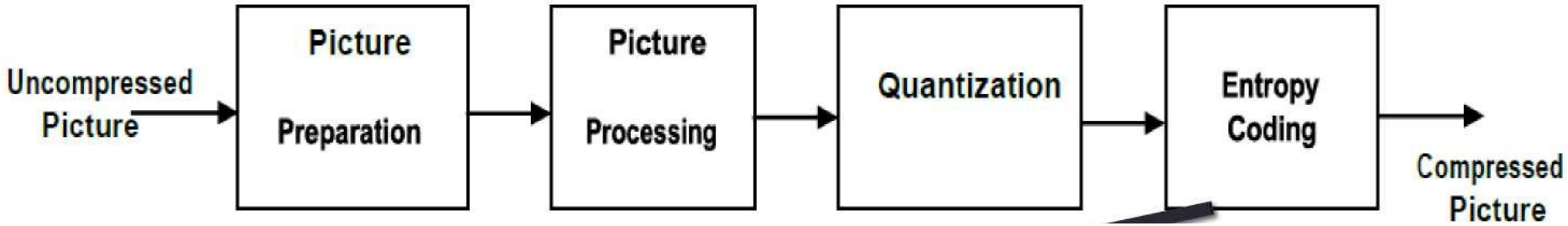
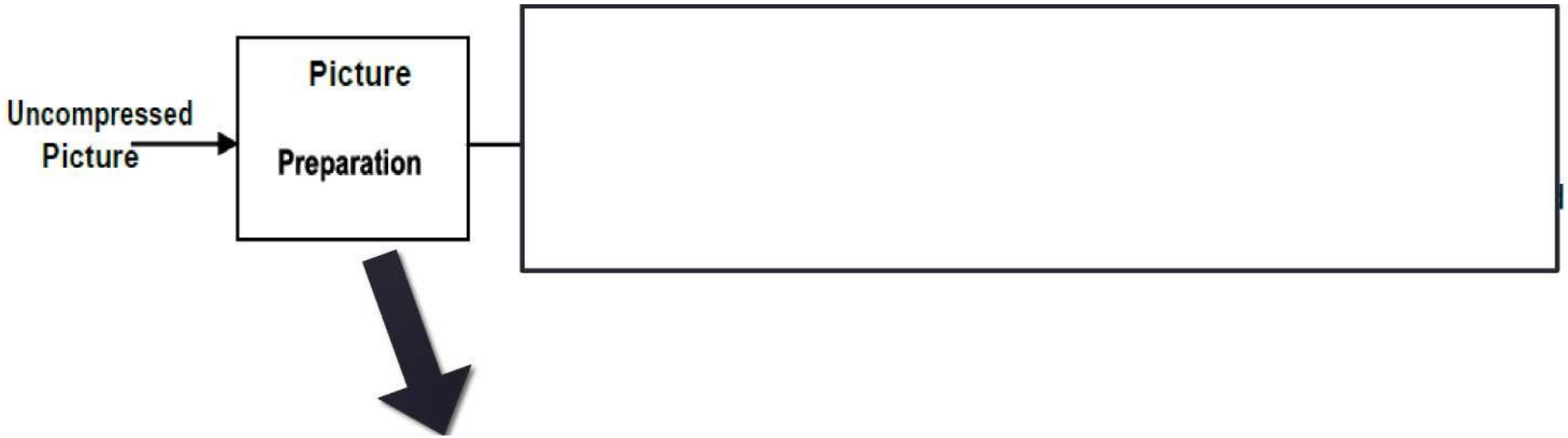


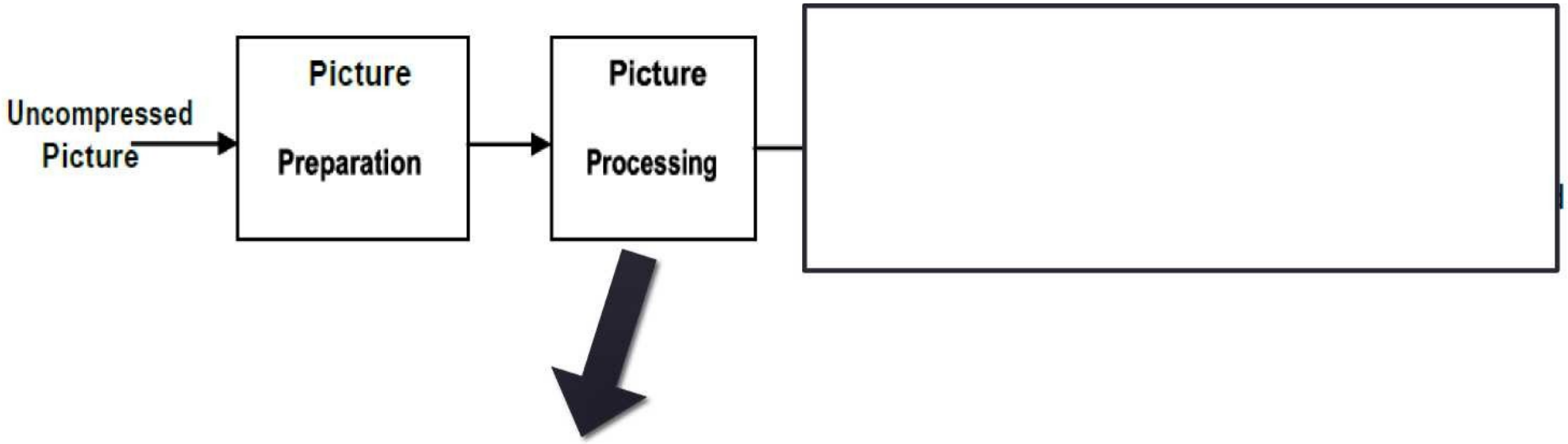

**Figure:** Major steps of data compression

# Basic Data Compression Techniques



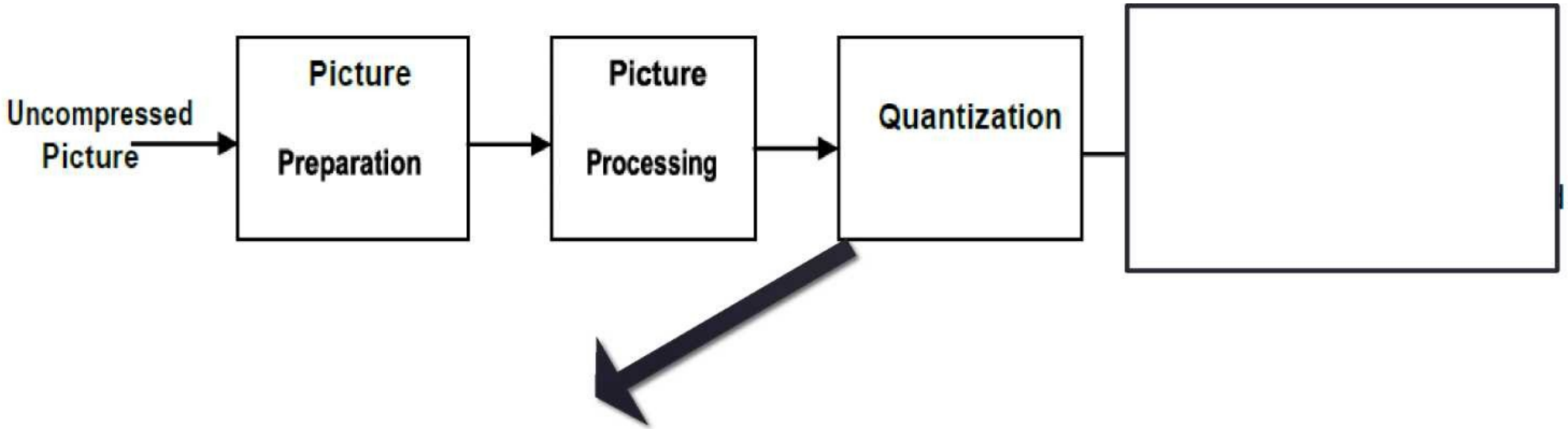

- Picture preparation: this includes analog to digital conversion and generating a appropriate digital representation of the information. An image is divided into block of 8 X 8 pixels and represented by fixed number of bits/pixel

# Basic Data Compression Techniques



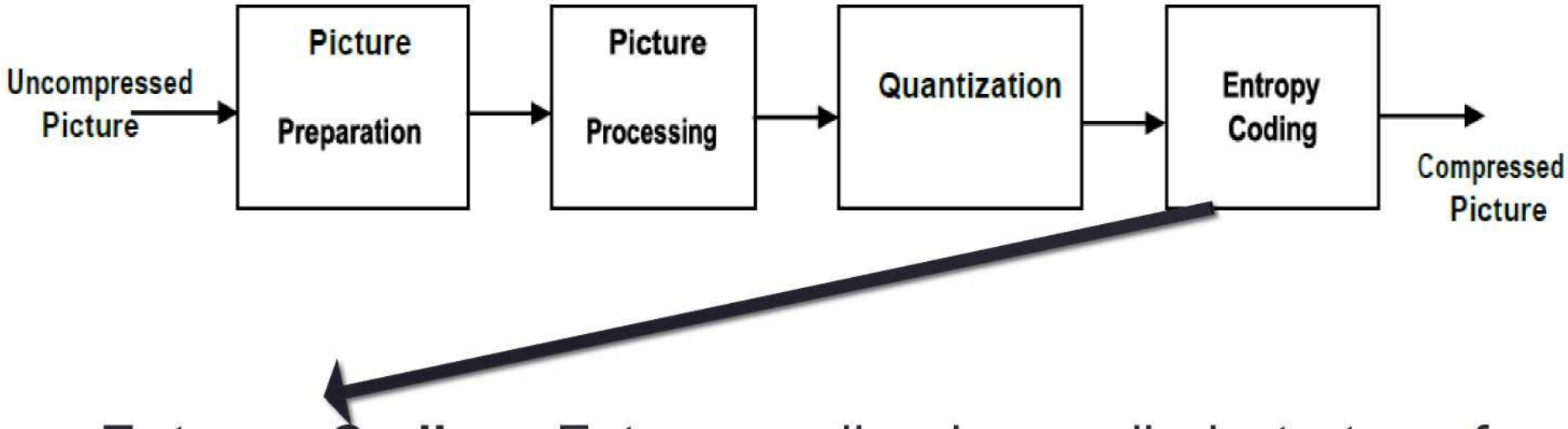

- Picture processing: Processing is the actually the first step of compression process which makes use of sophisticated algorithm of transformation from time to frequency domain. It can be performed using DCD (Discrete Cosine transform).

# Basic Data Compression Techniques




- Quantization Process: Quantization process the result of the previous step. It specifies the mapping of the real number into integer. This can be considered as the equivalent of p-law and A- law which apply the audio data. Quantization is performed using the different number of bits per coefficient.

# Basic Data Compression Techniques




- **Entropy Coding:** Entropy coding is usually last step of data compression. It compresses a sequence of digital stream without loss. For e.g. a sequence of zero's in the data stream can be compressed by specifying the number of occurrence followed by zero it.

# Basic Data Compression Techniques

- The **processing** and **quantization** cab be repeated iteratively several times in feedback loop.
- The term spatial domain (time domain) refer to the image plane itself and approaches in this category are based on discrete manipulation of pixel in an image.
- After compression, the compressed video stream contains the specification of the image starting point and an identification of the compression technique may be the part of the data stream. The error correction code may also be added to the stream.
- Decompression is the inverse process of compression.

## Data Compression and Coding Standards:

**JPEG (Joint Photographic Experts Group)**

The JPEG standard for compressing continuous -tone still pictures (e.g. photographs) was developed by photographic experts working under the joint auspices of ITU, ISO and IEC.

JPEG is significant in compression because MPEG or the standard for motion picture compression is just the JPEG encoding applied to each frame separately.

# Data Compression and Coding Standards:

**JPEG (Joint Photographic Experts Group) (cont..)**

There are some requirements of JPEG standard and they are:

- The JPEG implementation should be independent of image size.
- The JPEG implementation should be applicable to any image and pixel aspect ratio.
- Color representation itself should be independent of the special implementation.
- Image content may be of any complexity, with any statistical characteristics.
- The JPEG standard specification should be state of art (or near) regarding the compression factor and achieved image quality.
- Processing complexity must permit a software solution to run on as many available standard processors as possible. Additionally, the use of specialization hardware should substantially enhance image quality.

# Data Compression and Coding Standards:

## Steps in JPEG Compression

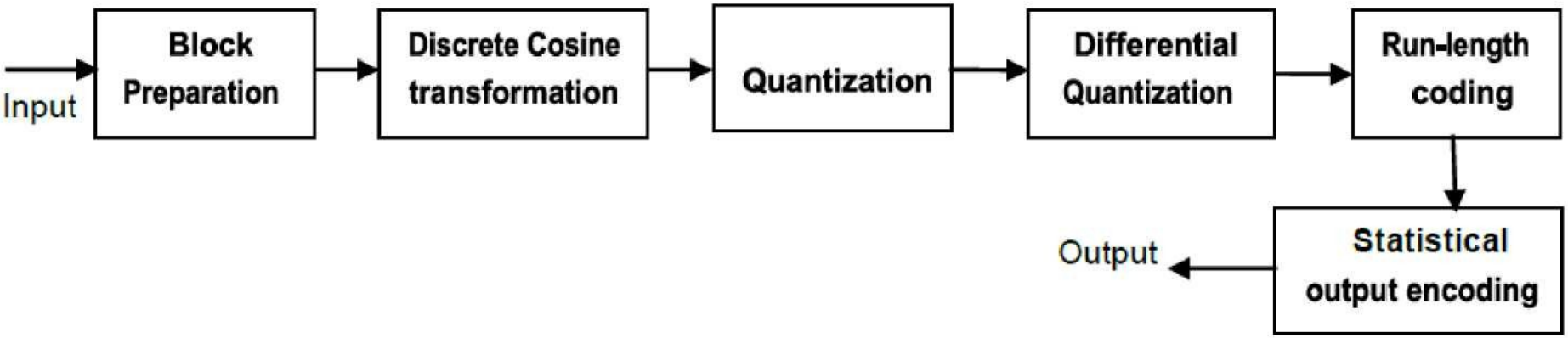



**Figure:** The operation of JPEG in lossy-sequential mode.

# Data Compression and Coding Standards:

## Steps in JPEG Compression (cont..)

### *Step 1: (Block Preparation)*

This step involves the block preparation. For e.g. let us assume the input to be 640*480 RGB image with 24 bits/pixel. The luminance and chrominance component of the image is calculated using the YIQ model for NTSC system.

$$Y=0.30R + 0.59G + 0.11B$$

$$I= 0.60R - 0.28G - 0.32B$$

$$Q= 0.21R - 0.52G + 0.31B$$

# Data Compression and Coding Standards:

## Steps in JPEG Compression (cont..)

### *Step 1: (Block Preparation) (cont..)*

For PAL system YUV model is used. Separate matrices are constructed for Y, I and Q each elements in the range of 0 and 255.

The square blocks of four pixels are averaged in the I and Q matrices to reduce them to **320*240**. Thus the data is compressed by a factor of two.

Now, 128 is subtracted from each element of all three matrices to put 0 in the middle of the range. Each image is divided up into 8*8 blocks. The Y matrix has 4800 blocks; the other two have 1200 blocks.

# Data Compression and Coding Standards:

## Steps in JPEG Compression (cont..)

### *Step 2: (Discrete Cosine Transformation)*

***Discrete Cosine Transformation*** is applied to each 7200 blocks separately.

The output of each DCT is an 8*8 matrix of DCT coefficients.

DCT element (0,0)is the average value of the block. The other element tells how much spectral power is present at each spatial frequency.

# Data Compression and Coding Standards:

## Steps in JPEG Compression (cont..)

***Step 3: (Quantization)***

In this step the less important DCT coefficients are wiped out.

This transformation is done by dividing each of the coefficients in the 8*8 DCT matrix by a weight taken from a table.

If all the weights are 1 the transformation does nothing however, if the weights increase sharply from the origin, higher spatial frequencies are dropped quickly.

# Data Compression and Coding Standards:

## Steps in JPEG Compression (cont..)

***Step 4: (Differential Quantization)***

This step reduces the (0,0) value of each block by replacing it with the amount it differs from the corresponding element in the previous block.

Since these elements are the averages of their respective blocks, they should change slowly, so taking the differential values should reduce most of them to small values.

The (0,0) values are referred to as the DC components; the other values are the AC components.

# Data Compression and Coding Standards:

**Steps in JPEG Compression (cont..)**

***Step 5: (Run length Encoding)***

This step linearizes the 64 elements and applies run-length encoding to the list.

In order to concentrate zeros together, a zigzag scanning pattern is used.

Finally run length coding is used to compress the elements.

# Data Compression and Coding Standards:

## Steps in JPEG Compression (cont..)

***Step 6: (Statistical Encoding)***

Huffman encodes the numbers for storage or transmission, assigning common numbers shorter codes than uncommon ones.

JPEG produces a 20:1 or even better compression ratio. Decoding a JPEG image requires running the algorithm backward and thus it is roughly symmetric: decoding takes as long as encoding.

# Data Compression and Coding Standards:

**Lossy Sequential DCT-based Mode of JPEG *Image Processing***

It basically involves the block preparation where the image samples are grouped into 8*8 pixels and passed to the encoder.

Then Discrete Cosine Transformation is applied to the blocks where the pixel values are shifted into the range [-128,127] with zero as the center.

Each of these values is then transformed using ***Forward DCT (FDCT).*** DCT is similar to ***Discrete Fourier Transformation*** as it maps the values from the time to the frequency domain.

# Data Compression and Coding Standards:

## Lossy Sequential DCT-based Mode of JPEG

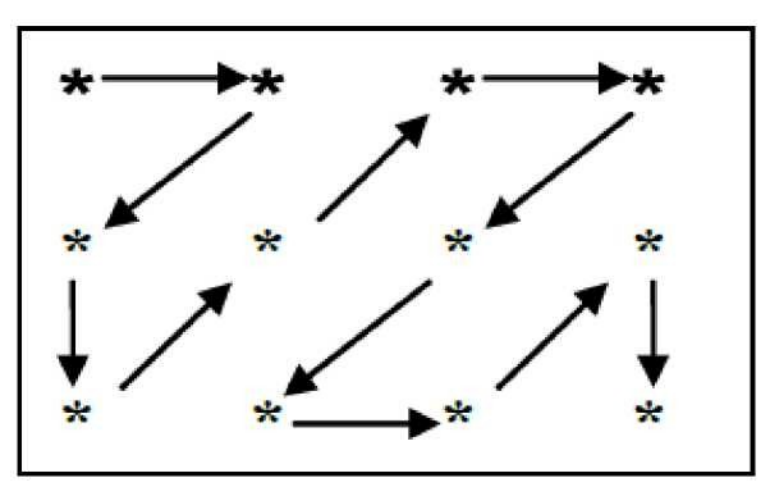
***Quantization***

The JPEG application provides a table of 64 entries. Each entry will be used for the quantization of one of the 64 DCT-coefficients

# Data Compression and Coding Standards:

**Lossy Sequential DCT-based Mode of JPEG *Entropy Encoding***

During the initial step of entropy encoding, the quantized DC-coefficients are treated separately from the quantized AC-coefficients.



# Data Compression and Coding Standards:

## Lossy Sequential DCT-based Mode of JPEG *Entropy Encoding*

- The DC-coefficient determines the basic color of the data units.
- The DCT processing order of the AC coefficients involves the zigzag sequence to concentrate the number of zeros.

JPEG specifies Huffman and arithmetic encoding as entropy encoding methods. However, as this is lossy sequential DCT-based mode, only Huffman encoding is allowed. In lossy sequential mode the framework of the whole picture is not formed but parts of it are drawn i.e. sequentially done.

# Data Compression and Coding Standards:

## Expanded Lossy DCT-based Mode

It differs from the sequential mode in terms of number of bits per sample. Here 12 bits along with 8 bits per sample can be used.

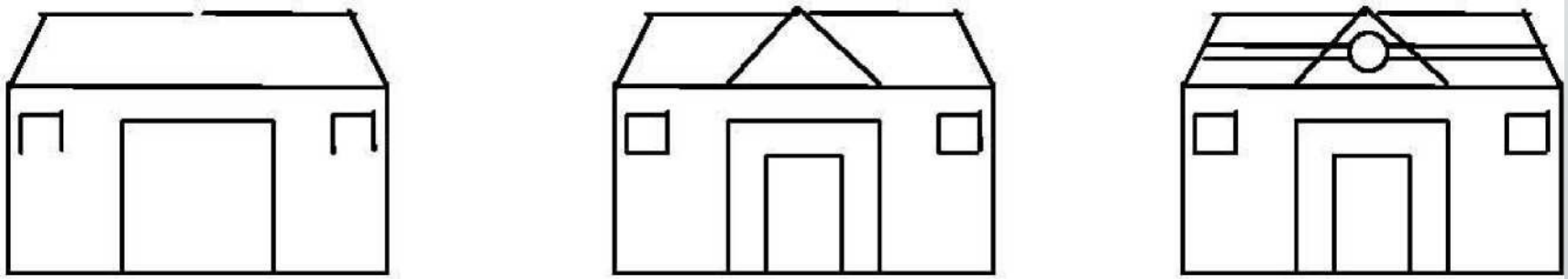


Figure: Progressive Picture Presentation

# Data Compression and Coding Standards:

## Expanded Lossy DCT-based Mode (cont..)

For the expanded lossy DCT-based mode, JPEG specifies ***progressive encoding*** in addition to sequential encoding.

At first, a very rough representation of the image appears which is progressively refined until the whole image is formed. This progressive coding is achieved by layered coding.

# Data Compression and Coding Standards:

## Expanded Lossy DCT-based Mode

Progressiveness is achieved in two different ways:

- By using a ***spectral selection*** in the first run only, the quantized DCT coefficients of low frequencies of each data unit are passed in the entropy encoding. In successive runs, the coefficients of higher frequencies are processed.
- ***Successive approximation*** transfers all of the quantized coefficients in each run, but single bits are differentiated according to their significance. The most-significant bits are encoded first, then the less-significant bits.

## Data Compression and Coding Standards:

### Hierarchical Mode

This mode uses either the lossy DCT-based algorithms or the lossless compression technique. The main feature of this mode is the encoding of an image at different resolutions, i.e., the encoded data contains images at several resolutions. This involves the sampling of images at higher resolutions at first and then subsequently reducing the horizontal and vertical resolution. The compressed image is subtracted from the previous result. The process is repeated until the full resolution is reduced considerably and no further compression is possible.

This requires more storage but as images at different resolution are available the systems using low resolution need not decode the image as the low resolution image is easily available.

# Data Compression and Coding Standards:

**MPEG (Motion Picture Expert Group)**

The Motion Pictures Expert Group was formed by the ISO to formulate a set of standards relating to a range of multimedia applications that involve the use of video with sound.

The coders associated with the audio compression part of these standards are known MPEG audio coders and a number of these use perceptual coding.

MPEG can deliver a data rate of at most 1856000 bits/second, which should not be exceeded. Data rates for audio are between 32 and 448 Kbits/second; this data rate enables video and audio compression of acceptable quality.

# Data Compression and Coding Standards:

## MPEG (Motion Picture Expert Group) (cont..)

The MPEG standard exploits the other standards and they are

- ✓ JPEG:

  JPEG is used as the motion picture is a continuous sequence of still image.

- ✓ H.261:

  H.261 video compression standard has been defined by the ITU-T for the provision of video telephony and videoconferencing services over an Integrated Services Digital Network (ISDN). It specifies two resolution formats with an aspect ratio of 4:3 are specified. *Common Intermediate Format* (CIF) defines a luminance component of 288 lines, each with 352 pixels. The chrominance components have a resolution with a rate of 144 lines and 176 pixels per line. *Quarter CIF* (QCIF) has exactly half of the CIF resolution i.e., 176*144 pixels for the luminance and 88*72 pixels for the other components.

# Data Compression and Coding Standards:

## MPEG (Motion Picture Expert Group) (cont..)

MPEG video uses video compression algorithms called ***Motion-Compensated Discrete Cosine Transform*** algorithms. The algorithms use the following basic algorithm

- ***Temporal Prediction:*** It exploits the temporal redundancy between video pictures.
- ***Frequency Domain Decomposition:*** It uses DCT to decompose spatial blocks of image data to exploit statistical and perceptual spatial redundancy.
- ***Quantization:*** It reduces bit rate while minimizing loss of perceptual quality.
- ***Variable-length Coding:*** It exploits the statistical redundancy in the symbol sequence resulting from quantization as well as in various types of side information.

# Data Compression and Coding Standards:

## MPEG (Motion Picture Expert Group) (cont..)

As far as audio compression is concerned the time-varying audio input signal is first sampled and quantized using PCM, the sampling rate and number of bits per sample being determined by the specific application.

The bandwidth that is available for transmission is divided into a number of ***frequency sub bands*** using a bank of ***analysis function*** which because of their role, are also known as ***critical-band filters.***

# Data Compression and Coding Standards:

## Video Encoding

Video is nothing but simply a sequence of digitized pictures. The video that MPEG expects to process is composed of a sequence of frames or fields of luma and chroma.

***Frame-Based Representation:***

MPEG-1 is restricted to representing video as a sequence of frames. Each frame consists of three rectangular arrays of pixels, one for the luma (Y, black and white) component, and one each for the chroma (Cr and Cb, color difference) components. The chroma arrays in MPEG-1 are sub-sampled by a factor of two both vertically and horizontally relative to the luma array.

***Field-Based Representation:***

MPEG-2 is optimized for a wider class of video representations, including, most importantly, field-based sequences. *Fields* are created by dividing each frame into a set of two interlaced fields, with odd lines from the frame belonging to one field and even lines to the other. The fields are transmitted in interlaced video one after the other, separated by half a frame time.

# Data Compression and Coding Standards: Video

## Encoding (cont..)

MPEG provides 14 different image aspect ratios per pixel which are coded in the data stream. The image refresh frequency is also encoded in the data stream. Eight frequencies are defined: 23.976Hz, 24 Hz, 25 Hz, 29.97 Hz, 30 Hz, 50 Hz, 59.94 Hz and 60 Hz.

# Data Compression and Coding Standards:

## Types of Image Frames in MPEG

There are two basic types of compressed frame: those that are encoded independently and those that are predicted.

The first are as ***intracoded frames*** or ***I frames.*** In practice, there are two types of predicted frames: ***predictive*** or ***P-frames*** and ***bidirectional*** or ***B-frames*** and because of the way they are derived, the latter are also known as ***intercoded*** or ***interpolation frames.***

# Data Compression and Coding Standards:

## Types of Image Frames in MPEG

***I-frames (Intra coded Images):***

**I- frames are encoded without reference to any other frames.**

**Each frame is treated as a separate** (digitized) **picture** and the Y, C , and C matrices are encoded independently using the JPEG algorithm.

b r The level of compression obtained with I-frames is relatively small. I-frames must be present in the output stream at regular intervals in order to allow for the possibility of the contents of an encoded I-frame being corrupted during transmission. The number of frames/pictures between successive I-frames is known as a *group of pictures* or *GOP.*

# Data Compression and Coding Standards:

## Types of Image Frames in MPEG

***P-frames (Predictive-Coded Frames):***

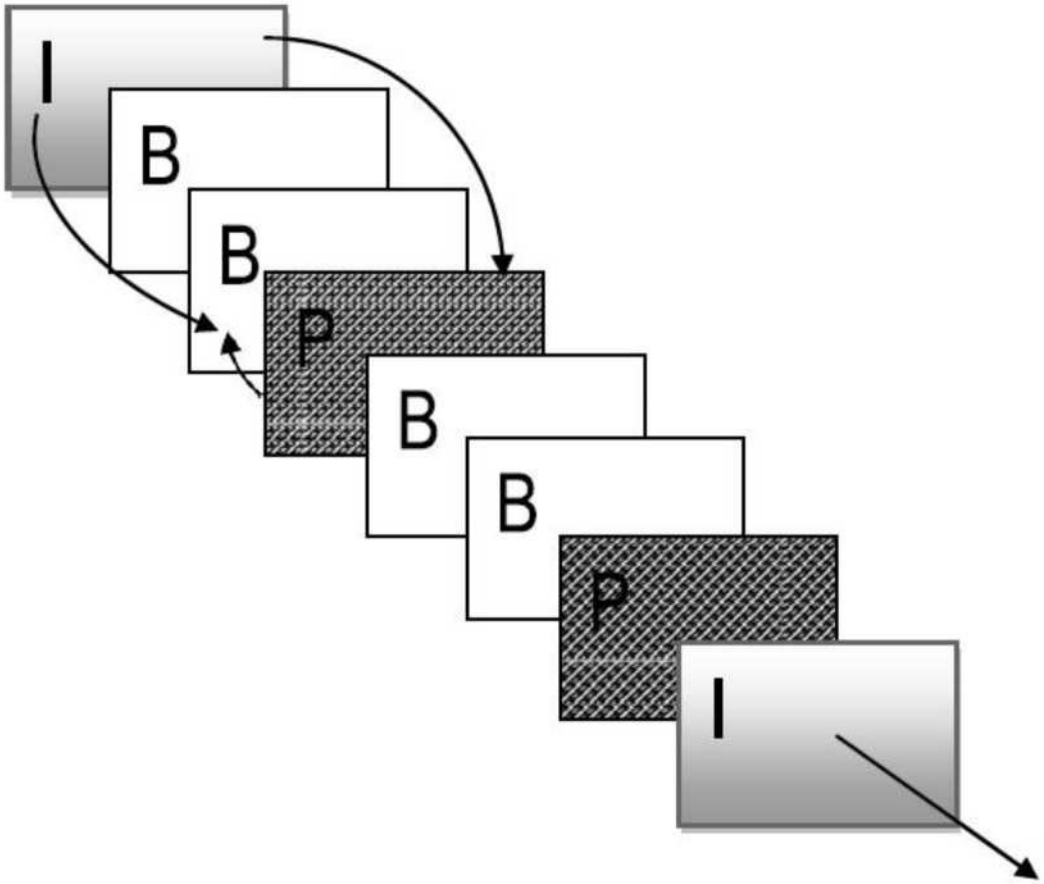
The encoding of a P-frame is relative to the contents of either a preceding I-frame or a preceding P frame.

As indicated, P-frames are encoded using a combination of motion estimation and motion compensation and hence significantly higher levels of compression can be obtained.

In practice, the number of P-frames between each successive pair of I-frames is limited since any errors present in the first P-frame will be propagated to the next. The number of frames between a P-frame and the immediately preceding I- or P-frame is called the *prediction span.* It is given the symbol M and typical values range from 1 through 3.

# Data Compression and Coding Standards:

## Types of Image Frames in MPEG (cont..)



# Data Compression and Coding Standards:

## Types of Image Frames in MPEG (cont..)

***B-frames (Bi-directionally predictive-coded frames):***

Motion estimation involves comparing small segments of two consecutive frames for differences and, should a difference be detected, a search is carried out to determine to which neighboring segment the original segments has moved. In order to minimize the time for each search, the search region is limited to just a few neighboring segments. Some applications may involve very fast moving objects; however, it is possible for a segment to have moved outside the search region. To allow for this possibility, in applications such as movies, in addition to P-frames, second types of frames are used called B-frames.

# Data Compression and Coding Standards:

## Types of Image Frames in MPEG (cont..)

***D-frames (DC-Coded Frames):***

D-frame has been defined for use in movie/video-on-demand applications.

D-frames are used for display in fast-forward or fast-rewind modes. D-frames are inserted at regular intervals throughout the stream. These are highly compressed frames and are ignored during the decoding of P- and Bcompression algorithm, the DC coefficient associated with each 8 * 8 block pixels- both for the luminance and the two chrominance signals- is the mean of all the values in the related block. Hence by using only encoded DC coefficients of each block of pixels in the periodically inserted D-frames, a lowresolution sequence of frames is provided each of which can be decoded at the higher speeds that are expected with the rewind and fast-forward operations.

# Data Compression and Coding Standards:

## Audio Encoding:

The time-varying audio input signal is first sampled and quantized using PCM, the sampling rate and number of bits per sample being determined by the specific application. The bandwidth that is available for transmission is divided into a number of *frequency sub bands* using a bank of *analysis filters* which, because of their role, are also known as *critical-band filters.*

Each frequency subbands is of equal width and, essentially, the bank of filters maps each set of 32 PCM samples into an equivalent frequency samples, one per subband, Hence each is known as a *sub band sample* and indicates the magnitude of each of the 32 frequency components that are present in a segment of the audio input signals of a time duration equal to 32 PCM samples.

# Data Compression and Coding Standards:

## Audio Encoding:

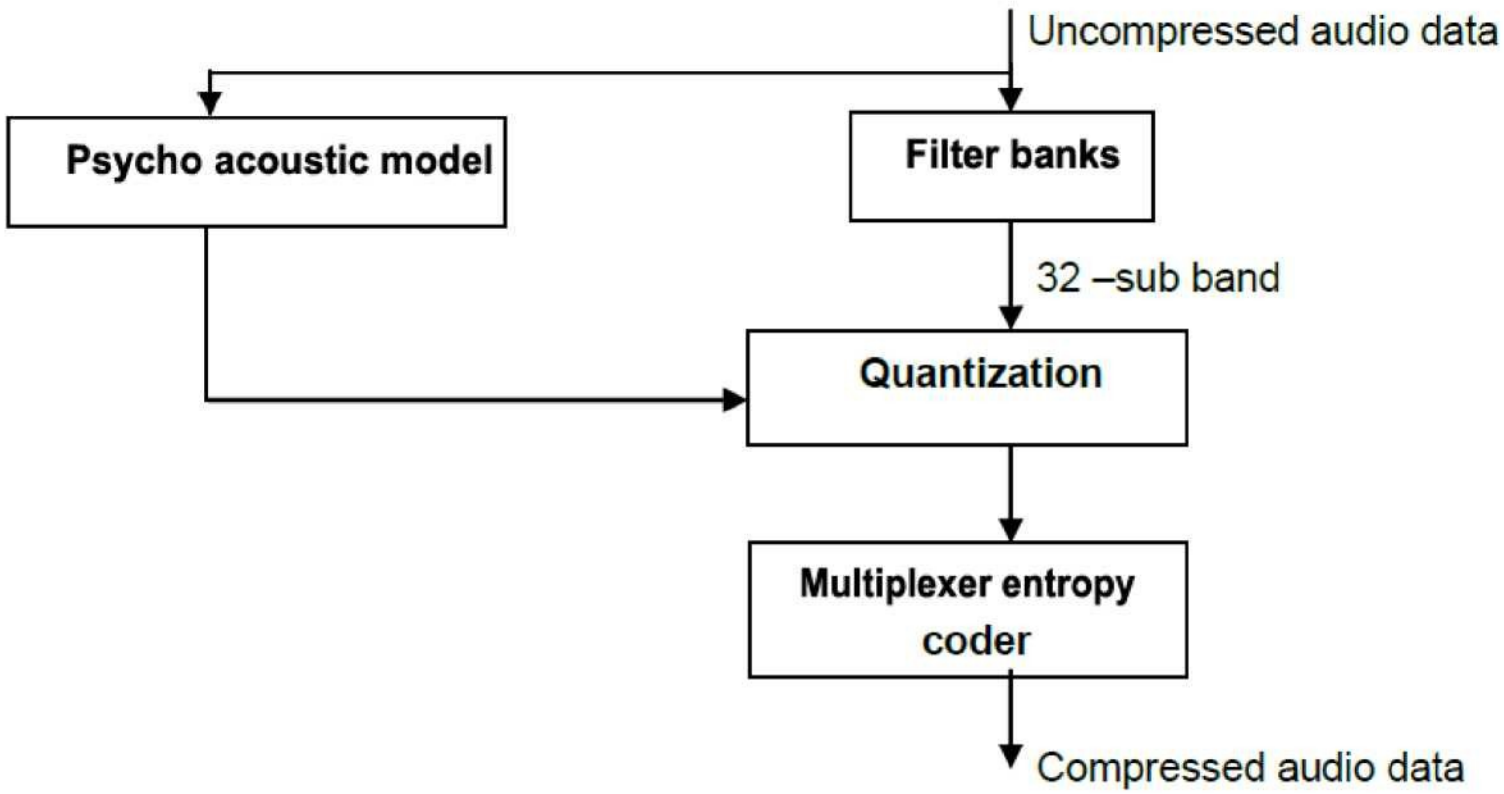



**Figure:** MPEG basic steps for audio encoding

# Data Compression and Coding Standards:

## DVI (Digital Video Interactive)

DVI is a technology that includes coding algorithms. The fundamental components are a VLSI chip set for the video subsystem, a well specified data format for audio and video files, an application user interface to the audio-visual kernel and compression, as well as decompression, algorithms. For encoding audio standard signal processor is used. Processing of images and video is performed by a video processor.

# Data Compression and Coding Standards:

**DVI (Digital Video Interactive)**

**Audio and Still Image encoding:**

Audio signals are digitized using 16-bits per sample. Audio signals may be PCM-encoded or compressed using the adaptive differential pulse coded modulation (ADPCM) technique. Supported sampling frequencies are: 11025Hz, 22050Hz and 44100 Hz for one or two PCM-coded channels. And 8268Hz, 31129Hz, 33075Hz for ADPCM.

# Data Compression and Coding Standards:

**DVI (Digital Video Interactive)**

**Audio and Still Image encoding:**

For Still Images, DVI assumes an internal digital YUV format for image preparation. Any video input signal must first be transformed into this format. The color of each pixel is split into luminance component and the two chrominance components (U and V). The luminance represents the gray scale image. With RGB, DVI computes the YUV signal using the following relationship.

$$Y=0.30R+0.59G+0.11B$$

$$U=B-Y$$

$$V=R-Y$$

# Data Compression and Coding Standards:

**DVI (Digital Video Interactive)**

**Audio and Still Image encoding:**

It leads to:

$$U=-30R-0.59G+0.89B$$

$$V=0.70R-059G-0.11B$$

DVI Determines the components YUV according to the following:

$$Y=0.299R+0.587G+0.144B+16$$

$$U=0.577B-0.577Y+137.23$$

$$V=0.730R-0.730Y+139.67$$

# Data Compression and Coding Standards:

## DVI (Digital Video Interactive)

## Audio and Still Image encoding:

DVI is able to process image in the 16-bit YUV format and the 24-bit YUV format. The 24-bit YUV format uses 8 bits for each component. The 16-bit YUV format coded the Y components of each pixel with 6 bits and the color difference components with 5 bits each.

There are 2 bitmap formats: Planer and Packed.

***Planer:***

All data of the Y component are stored first, followed by the U component values and then all V values.

***Packed:***

For the packed bitmap format, the Y, U, and V information of each pixel is stored together by the data of the next pixel.

# 4. Data Compression Methods

**4.1. Coding Requirement**

**4.2. Types of Compression:**
Lossy and LosslessCompression

4.2.1. Run length Coding, Entropy Coding

4.2.2. Huffman Coding

4.2.3. Arithmetic Coding

**4.3. Coding Standards**

4.3.1. JPEG: JPEG compression steps

4.3.2. MPEG: Evolution of MPEG, Data Stream of MPEG,MPEG Audio

**4.4. Digital Video Interactive (DVT), h.261(p✖64)**

# Assignments VI

(1) How is source coding different from entropy encoding? Describe about the MGPEG video compression.

(2) What are the different types of compression technique used? Explain in detail any one Source encoding technique used for data compression.

(3) How is source coding different from entropy encoding? Describe about the JPEG compression.

(4) What is data compression? Why multimedia data should be compressed? Describe the JPEG compression with its different modes.